

Highlights

A Comprehensive Highway Benchmark for Investigating State Representations in Reinforcement Learning for Lane Changing

Meriem Bouali, Abderrazak Sebaa, Nadir Farhi

- Built highway benchmark covering a wide range of driving scenarios and road layouts.
- Proposition of a new environment state that is not limited by the number of lanes or road geometry.
- First investigation of state encoding impact on reinforcement learning for lane changing.

A Comprehensive Highway Benchmark for Investigating State Representations in Reinforcement Learning for Lane Changing

Meriem Bouali^{a,*}, Abderrazak Sebaa^a, Nadir Farhi^b

^a*Laboratoire LITAN, École supérieure en Sciences et Technologies de l'Informatique et du Numérique, RN 75, Amizour 06300, Bejaia, Algérie*

^b*Cosys-Grettia, University Gustave Eiffel, F-77454 Marne-la-Vallée, France*

Abstract

Autonomous vehicles have garnered significant attention in recent years due to their potential to improve road safety and efficiency. In this research, we propose an enhanced driving model specifically designed for highway scenarios, with a particular focus on lane-changing maneuvers. In this research, our objective is improving driving models specifically designed for highway scenarios, with a particular focus on lane-changing maneuvers.

Keywords: Lane Changing, Reinforcement Learning, Autonomous vehicles, Sequential decision-making.

1. Introduction

2. Results and Discussion

3. Conclusion

*Corresponding author

Email addresses: `bouali@estin.dz` (Meriem Bouali), `sebaa@estin.dz` (Abderrazak Sebaa), `nadir.farhi@univ-eiffel.fr` (Nadir Farhi)